

Inequalities

Worksheet 1 (Non-Calculator)

1. S15/qp11/q8

Find the integers n that satisfy $20 < 4n - 3 < 30$. [2]

2. S15/qp12/q23(a)

Solve the inequalities $-4 \leq 2x - 5 < 7$. [2]

3. S16/qp12/q13(b)

Write down one possible value of x that satisfies each inequality.

i. $2 < \sqrt{x} < 3$ [1]

ii. $-1 < x^3 < 0$ [1]

4. S17/qp12/q9

a. Write down all the integers that satisfy the inequality $-\frac{3}{2} \leq x < 2$ [1]

b. Complete the following inequality with a fraction. [1]

$$\frac{3}{4} > \dots > \frac{1}{2}$$

c. Write down an irrational value of n that satisfies this inequality. [1]

5. W17/qp11/q9

a. Solve the inequality $12 - 2x < x$. [1]

b. Find the integer values of n that satisfy $-8 < 2n \leq -4$. [1]

6. W17/qp12/q14

Find the two solutions of $\frac{x}{3} - 1 < \frac{3x}{4}$ which are negative integers. [3]

7. W20/qp21/q6

Solve the inequality $6x - 7 > 5 - 2x$. [2]

Answer Key:

1. 6,7,8

2. $-\frac{1}{2} \leq x < 6$

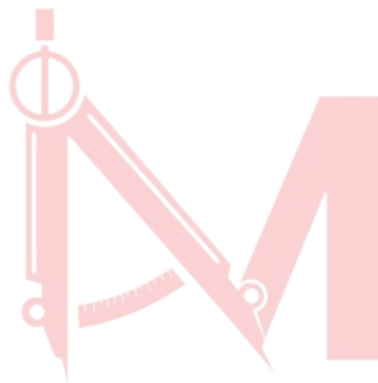
3. any value in range $4 < x < 9$ any value in range $-1 < x < 0$

4. a. -1,0,1 (b) correct fraction (c) irrational number between 2 and 3

5. $x > 4$ (b) -3 and -2

6. -2 and 1

7. $x > 1.5$



Mathlete
= by Saad